Did the Role Fire? Manipulation Checks for Role-Prompted Multi-Agent LLM Systems

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Abstract

Many multi-agent LLM papers claim that the improvements they observe follow from the transformations implied by the role prompts (verifier, critic, devil's advocate) their agents use, often without checking whether these prompts implement them. We propose executable manipulation checks, drawing on the manipulation check paradigm in experimental psychology: predeclared behavioral checks, compared to a length-matched neutral baseline.

We test six role prompts in three versions each: an explicit implementation, a bare role-name, and a length-matched neutral prompt. Of six diagnostic checks on 200 examples each, one meets the pre-set statistical criteria, one reverses, and three show no difference from neutral; we performed a sixth check that was non-diagnostic after audit.

In terms of raw accuracy on the 1,000 scored HaluEval items, the three versions of the council vary within a single percentage point. None shows a significant effect on leaderboard score, according to our pre-registered test, and all score lower than a single call to the model. However, bare role-name prompts do show major reductions in refusal rates, suggesting that role prompting may affect refusal behavior, even if it doesn't implement its specified manipulation.

We show silent failures in some of the coalition attribution analysis used in this literature, on simulated records. Retraining ablations may produce a Shapley value of zero that still meets efficiency. Masking approximations may also not be ranking-preserving. A post-collection audit further shows that manipulation checks themselves need degeneracy diagnostics: of six pre-registered checks, five included specific guards; two were defeatable, three didn't vary on observed support, and the one check that passed had none. Manipulation checks should survive executions.

1 Introduction

Multi-agent LLM pipelines often give their agents distinct tasks (e.g. critic, verifier, or devil's advocate), and attribute observed downstream gains to the corresponding information processing event. However, the information processing event is often not directly validated as having occurred: the downstream metric alone is not sufficient to validate its existence, since downstream results are a function of all aspects of the prompt, including the defined information processing event. Prior work shows that LLMs better identify errors in others' text as compared to their own [9, 31], but that only justifies adding a second agent, not a particular persona.

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In the experimental sciences, it is common to validate whether or not an intervention occurred via *manipulation checks*. If one cannot validate that their intervention occurred, then there is no reason to believe they have measured the intervention. We introduce the manipulation check to the evaluation of multi-agent LLM pipelines. Our goal is only to transfer the tool to LLM research, not to assert its primacy: §2 names the closest existing work, and Appendix C states what our search covered and what it did not.

In our work we observe this channel directly: an empty role label prompt refuses at a statistically significant lower rate versus either the entire task or the length-controlled version, with no task performed. If a model behaves differently when asked different questions, there will be a change in score independent of whether the task was performed; in this case, the difference lies on un-scored items, but we've still seen the mechanism in action (if not its downstream consequences).

For this, we use six functional roles (each with a unique information processing operation) drawn from a proposed council design. While the roles are motivated by a computational theory of cognition [24], they are simply the *subject of our analysis, not our point* — we do not endorse the council design. We find that role prompts mostly do not do what they promise: one in six roles passes its manipulation check, one role changes the score significantly in the *opposite* direction of their stated function, and three are not distinguishable from a length-matched control with no prescribed operation, with one non-diagnostic after audit.

Contributions. **(i) Manipulation Checks for Role Prompting Agents**: an implementable and pre-registered role manipulation check (compared against a length-matched neutral control), for six roles, five of which were pre-registered with an explicit anti-degeneracy term. Post-collection stress tests show that the presence of such a term does not establish its effectiveness: three didn't vary on the observed support, and two admit simple degenerate solutions. **(ii) An Analysis of Manipulation-Check-Passing Rates vs. Score-Changing Rates**: six roles were categorized along both dimensions, by a pre-registered rule (the latter dimension is null). Again, the *point* is the check and criteria; the results are irrelevant. **(iii) A Multi-Family Model Analysis**: our manipulation checks analyzed against the same items in an entirely separate pretraining family — shows that while our overall conclusions remain the same, *one validated check does not transfer across the two families tested*, motivating our choice to perform the manipulation check, and *not* simply use a checklist. **(iv) A Pre-Registered, Auditable Experiment**: pre-registered SESOIs; pre-registered per-role predicted harms (and benefits); structural safeguards ensuring no fabricated data makes it into a results table; a first-class table of exclusions.[1]

2 Related Work

Testing role prompting assumptions. Studies show persona assignment is ineffectual at improving model accuracy, whereas *oracle* persona assignment is effective, suggesting the key is selection, not adoption [38]. Similarly, studies using expert personas have mixed effects across different tasks and outcomes [18]. Prompting models with different personalities alters behaviour, with performance depending on the composition of the agent group [37]. For this reason, there is reason for caution about role prompting, rather than reasons to ensemble with it [23]. Any role prompting improvement should be considered in this light.

Multi agent councils. Models are increasingly being ensembled across agents [4, 15, 20]. The most direct comparison is with *Council Mode* [35] which also shows reductions in hallucinations on the same two datasets as we use. Their method combines three different models from three different providers plus a specialised synthesising model; this means their improvement is a joint effect of different specialisms, diversity of lineage, and synthesisation capability, which can't disentangle each other. By contrast we keep the lineage constant and have no dedicated synthesiser, differentiating each arm based solely on the prompt. We also have reason to be pessimistic about council mode. The first is that deliberation may be an illusion [33]. Second is that correlated failure limits improvement from combination [8, 21].

[1] Code, prompts, collected records, both benchmark corpora and the pre-registration are available at `https://github.com/kevzho/role-prompted-systems`. `REPRODUCE.md` gives the exact command for each reported result; all of them run offline against the shipped records, with no API key and no network access.

Self correction and verification. Self-correcting models have been shown to have questionable improvements, with self-correction worsening answers [19], self-verification failing to identify model mistakes, and improvements being an illusion of evaluation [9, 31]. For this reason, we prefer to use objective and actionable critiques over self critiques.

Testing for role adherence, closest prior work. Many existing works evaluate whether an LLM "acts according to its assigned role" on a prompt; e.g. to what extent the LLM displays the *persona* on psychological scales through interview [34]; whether an LLM "sticks to" the *persona* when completing open-ended tasks [30]; whether *persona* parameters affect model predictions at all [17]; specific checklists of role adherence for *role-playing agents* [28]; "disobey role specification" as a specific failure mode for *multi-agent systems* extracted from execution traces [7].

We do not claim novelty, and there are two prior works sufficiently close to cite: Dietrich [11] construct role-specific tests of adherence to an advocate role using a stance classifier, to "explicitly test adherence to the respective roles;" this is as close as we can find to what we propose in a multi-agent LLM stack, and is a contemporaneous, non-peer-reviewed preprint, and we weight it accordingly. Bieberstein et al. [5] conduct an automatic "manipulation check" on a neutral prompt baseline *before* reporting results in a single-agent HCI study, which is identical to the procedure in psychology.

The novelty here does not lie in testing role adherence itself but in its application, in that the aforementioned tests a) are not declared in advance; b) lack a baseline or length-unmatched baseline; c) test adherence to a persona, not an explicitly-specified action; d) are never interpreted as a precondition to interpreting a **benchmark**.

Architectural novelty, baselines, and provenance. Much prior work explores structured agent architectures [32], including functional role agents governed by a routing agent [29] and role architectures inspired by clinical frameworks [36] (see also Lee et al. [24] for the related research question of how and when agents "spontaneously" implement such computations). We do not claim novelty on the agent design itself, but only on the quality control mechanism. Our baseline uses cost-matched pre-registration, common for routing experiments [27]; the results we obtain instead use a same-model single-call baseline (§4.2). The most promising hope for functionally constraining LLMs lies in steering at the activation level [3], although such experiments have proved unreliable [6]; we defer mechanistic intervention to future work. The archetypes we chose were initially inspired by prior work on computational theories of mind and information processing-accounts in clinical psychology and psychiatry [1, 2, 10, 13, 14, 22]: all such connections are limited to motivation of a functional role, and our archetypes are entirely specified, implemented, and assessed in functional terms, and none of our conclusions apply to any clinical group. *These roles are functional AI processing constraints. They are not diagnostic models, simulations of lived experience, or claims that mental health conditions confer a single cognitive mechanism.*

3 Executable Manipulation Checks

3.1 Giving f empirical content

The definition of an information processing agent is that it executes a manipulation $f(C_i, y, x)$ on y. There are no restrictions on f, meaning any post-hoc observation about a model's behavior given a certain prompt can be claimed to be implementing an IPA, which makes the definition unfalsifiable and purely descriptive. To address this, we will demand a mechanism to check that the claimed manipulation was indeed performed.

Definition 1 (executable manipulation check)**.** *An executable manipulation check for a role i is a scalar-valued function m_i, taking as input structured output records, specified prior to running the experiment, where a higher value of m_i indicates that the structured output records are consistent with the claimed manipulation performed by C_i. For a role's performance to be ascribed to its operation, its manipulation check m_i must discriminate between the structured output records of the role, and a length-matched control condition.*

Throughout, *implements* is operational: the role must produce the predeclared signature in its structured records, and no claim is made about latent computation that leaves no such trace.

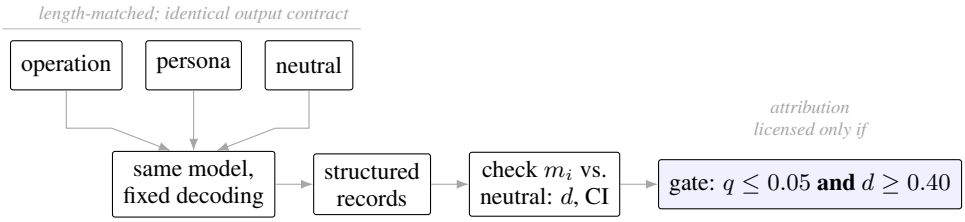


Figure 1: The check pipeline instantiated each role via three system prompts sharing one output contract (the predeclared processing instruction, the role label alone, and no instruction at all). All three traverse an identical path, so the contrast at the end isolates the prompt, and the gate decides whether a benchmark delta may be read as an effect of the operation at all.

In experimental psychology, a manipulation check can confirm that the experimental manipulation was conducted as planned, before making any conclusions about the experimental condition's effects [12]. They're common in published studies: Ejelöv and Luke [12] report their usage across 207 published experiments. In multi-agent LLM literature, they are rare; §2 names the closest cases.

While we use manipulation checks, we also acknowledge that their use in psychology is itself controversial, both because they can trigger self-reinforcing or self-defeating dynamic processes [16] and because the majority of checks reported do not account for these dynamics [12]. These concerns do not apply in this context, as our manipulation checks are ex-post functions over already generated structured outputs, that only use data produced by the benchmark. They are therefore non-interventionist, and do not 'ripple' into the experiment: the key concern with manipulation checks in human studies stems from a necessary human intervention, which is not required when the check is executable code.

3.2 Three prompt variants

We present three conditions: operation vs neutral, which is the key contrast, and a persona condition that disentangles the role condition from the operation it represents. We programmatically test that all control conditions are padded so their length is within 10% of the operation prompt (to ensure that only the operation is helpful, not longer token strings), that our persona conditions provably do not contain any words from the operation, and assert that none of the prompts (when rendered) contain any medical terms. In all cases, we use a standardized structured output format (a single JSON block with answer, abstain, confidence, list of atomic statements with whether they have evidence, counterevidence and failures), which is necessary to enable all conditions to be evaluated by the same set of metrics (the costs of this approach are discussed in §D).

3.3 The six checks

For each condition, we specify a manipulation check. These manipulation checks (rate of counterclaims and flags, number of contradictions found (corrected for abstention), number of atomic statements made, diversity $\times$ concentration of hypotheses, number of sourced and unsourced alternatives proposed and negative expected calibration error (NECE)) were all explicitly specified, prior to data collection, by condition, and are only properties of the transformation represented by the condition rather than the "correctness" of a solution (see Appendix I for details on how these checks are calculated).

3.4 Decision rule and gate

The checks are paired at the item level (the same item under all three variants) and per-item differences are bootstrapped over 2,000 resamples. Both conditions in Figure 1 gate on the number of observations, $n = 200$ — a tight interval around a negligible difference will pass a significance-only rule, and a large standardized effect with an interval spanning zero will not offer evidence for a difference. The threshold is on a standardized but not absolute scale because the six checks have no common unit (two return unbounded counts, one a rate difference in $[-1, 1]$, one a distinct-item

count). A single absolute threshold would be uninterpretable. Bounded rates are sensitive to their inputs; rounding errors can seem like large moves.

The check runner exits non-zero if any of the roles fail to pass their checks, making them a gate on a collection run instead of a report that will be produced and not read. If a role fails, it means that this role hasn't validated the predeclared behavioural signature of its operation — equivalently, no matter how good it scored on the benchmark, this delta isn't licensed as evidence for an effect of this operation.

4 Experimental Setup

4.1 Pre-registration

We registered hypotheses, effect sizes, arm definitions, decision rules, and per-role predictions before data collection. Two amendments were applied predating the data they're related to. Specifically, amendment A2 claimed that improvement on the manipulation check was our primary outcome, rather than the benchmark gain. A3 acknowledges a pilot which led to the analysis of the 47-item test split and pre-registered Benjamini–Hochberg control for the six role checks.

The registered decision rule is conjunctive. A role passes only if its paired operation-versus-neutral contrast survives FDR control at $q = 0.05$ across the six-role family *and* reaches $d \geq 0.40$; a role is recorded as moving a benchmark only above a leave-one-out contrast of 0.05. All intervals are 95% paired bootstrap intervals over 2,000 item-level resamples. The scoped study is underpowered for the original council-gain hypothesis by design — at $n = 1{,}000$ the detectable effect is ≈ 0.048 against a SESOI of 0.03 — so that hypothesis is reported as **not tested**, never as a null. The manipulation check is differently placed: at $n = 200$ paired, the minimum detectable effect at 80% power is ≈ 0.20, so the registered $d \geq 0.40$ threshold sits comfortably above the detection floor.

4.2 Arms and cost matching

The registered design has four arms: a full-operation council, a persona-label-only council, a length-matched neutral council, and a cost-matched single-agent baseline. The council arms differ only in the prompt (§3.2); model, decoding, output contract, and aggregation are identical.

The fourth arm was collected, but not as registered. The registered baseline was cost-matched. So, what we ran is the single-agent arm on the same model the councils use with one direct call, making it neither cost-matched nor call-matched to the councils. The preregistered cost-matched contrast remains uncollected. A council deficit against this baseline is therefore a deficit against something that also costs less: a stronger practical claim and a weaker answer to the registered question. §4.4 states the consequence for the attribution baseline.

Two benchmarks are collected, both materialised by script from pinned immutable revisions and verified by content hash so the corpus is reproducible byte-for-byte: HaluEval's QA split [25] at `207f479c`, expanded to 20,000 binary items, and TruthfulQA's generation split [26] at `741b8276`, 817 items. We use both for non-commercial research evaluation, following the terms stated in their licenses. No personal data is contained and we redistribute with the original license with attribution, with license identifiers and URLs recorded per benchmark in `data/materialize_manifest.json`.

Accuracy analysis is limited to HaluEval. Because the TruthfulQA generation split does not provide an answer index (grading uses a judge model), the gold answer index is necessarily undefined for all TruthfulQA questions. We classified TruthfulQA as free-form in preprocessing but did not recognise before collection that the registered accuracy analysis needs a gold answer index, which the generation split lacks. Examples from the TruthfulQA benchmark are, however, reflected in behavioural metrics shown in §5 (e.g., refusal, abstention rates). All accuracy metrics are, in turn, limited to HaluEval and reported as such (total examples), everywhere it appears. For the benchmark arms, we use 1,000 HaluEval examples per arm (see Appendix E).

4.3 Models

For our benchmark arms, we host `gpt-5.6-terra` in Microsoft Foundry (eastus2 Global Standard), and for cross family replication in §5.3 we host `grok-4.3` in the same. Our pre-registered cost-

matched arm uses `gpt-5.6-luna`, but as we switch Luna in the collected single agent arm to Terra (§4.2), results do not include Luna. For all arms and roles, we use `temperature=0`, 1,024 output tokens, `seed=1337`, `timeout=120s`. Of all provenance facts, only the following two are asserted (not inferred) — and only these two: (i) We specify the deployment name as the ID in our records. This deployment is known to have used model version `2026-07-09`. The deployment name does not reference this version: we thus specify the version in the manifest and do not attempt to infer it from the records themselves; (ii) We specify the serving route (vendor, region, deployment mode), not infer it, since our collection script accepts a provider argument. All other provenance facts are inferred (prompts via hash, simulated/reality from provider attestation, splits by content).

Compute. All model calls were hosted as API calls, without using any GPU cluster nor hardware on the provider's side. In collection, we issued 35,700 model calls. This includes three council arms at 1,500 items each $\times$ 6 different roles, another single agent arm at 1,500 items, and then three prompt conditions at 200 items $\times$ 6 roles in each of `gpt-5.6-terra` and `grok-4.3`. 35,208 (2.2% absent rate) returned usable records, with recorded usage totalling about 8.09M output tokens ($128.66). Each arm in the council costs six calls per item: around 3,350 input and 1,428–1,659 output tokens per item ($0.0239–$0.0267), with mean per-agent latency at 3.1–4.5 s; the single-agent baseline averages 2.44 s (median 2.23 s) and $0.001792 per call. Per-run cost, token, and latency breakdowns are in each run's `evaluation.json` under `cost`. The synthesizer is the only trained component: 81–108 optimizer steps, 34.7–55.3 s wall clock per run on one Apple Silicon laptop (MPS, no CUDA). Every analysis in this paper re-runs from the shipped records in under a minute on the same machine, with no API access.

4.4 Attribution estimator

In this work, we specify the value of the empty coalition. As specified in our pre-reg, we define $v(\emptyset)$ as the direct answer base model evaluated on identical per-sample inputs, with records provided by the single-agent arms and joined into our council records. Note that we are able to join 1,134/1,500 records per arm (including all 1,000 scorable records): we do not infer missing values for 366 TruthfulQA records where the baseline abstained (or returned free-form text with no answer index), we simply report these as missing. (Either way, leave-one-out does not consider the empty coalition at all.)

For per-role attribution we perform leave-one-out by retraining on each coalition (seven trainings total rather than the 64 required for a full Shapley computation, which is computationally prohibitive at this scale; using a masking approximation to improve tractability deviates from retraining by more than the effect size of interest, see Appendix F). Explicitly, leave-one-out is not a Shapley value and we do not claim it to be: in particular, it is not efficient, so per-role contributions do not have to add up to $v(A) - v(\emptyset)$, and it attributes all role–council interactions to each role. Writing v in Harsanyi dividend form $v(S) = \sum_{\emptyset \neq T \subseteq S} d(T)$, leave-one-out evaluates to $v(A) - v(A \setminus \{i\}) = \sum_{T \ni i} d(T)$ while the Shapley value is $\sum_{T \ni i} d(T)/|T|$, so the two differ by $\sum_{T \ni i,\, |T| \geq 2} \frac{|T|-1}{|T|} d(T)$: the standalone term $d(\{i\})$ cancels and leave-one-out charges role i the whole of each interaction rather than its $1/|T|$ share. Substitutable roles are negative dividends, so under the pairwise substitutability registered in advance this difference is negative and every leave-one-out contrast is biased downward relative to the Shapley value. The sign is imported rather than measured — the seven coalitions evaluated here do not identify $d(T)$ for $|T| \geq 2$ (Appendix G) — but the bias runs against a positive verdict either way, so a role exceeding the 0.05 leave-one-out (LOO) threshold has passed a conservative test.

Provenance is derived rather than asserted. Refusals, truncations, malformed responses, and retries are counted and published per role rather than imputed. In the pilot and manipulation-check runs reported here, those rates are zero across 1,128 pilot calls and the 3,600 calls; the benchmark arms refuse only on TruthfulQA items, entering no accuracy contrast (Appendix H). Appendix B gives the mechanism.

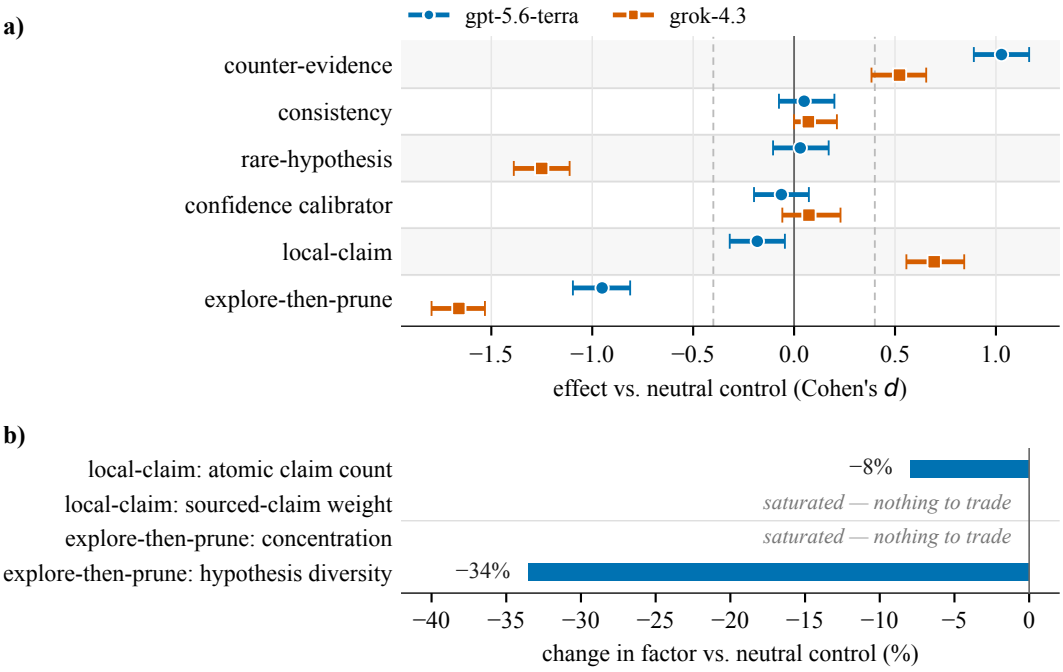


Figure 2: (a) Standardized difference of stated role effect against length-matched neutral condition, on 200 datapoints, in two pretraining families, with 95% bootstrap confidence intervals. Dashed lines at pre-registered $|d| \geq 0.40$. (b) These flips are not tradeoffs because both measures are product rules, with the factor being traded against pinned to maximum in both arms of the experiment, leaving all the action to the factor we told it to increase.

5 Results

5.1 Did the roles fire?

In Figure 2 we perform the check for each role against the length-matched neutral condition on 200 development items, in a paired bootstrap with 2,000 resamples (and no missing data, across all 3,600 invocations). **Only one in six roles accomplishes its stated operation:** counter-evidence checking more than doubles the rate of counterclaims and flagging relative to a length-matched neutral instruction ($d = 1.03$). This check is a sum of two components and they do not move together: explicit counterevidence items rise ($d = +1.48$ here, $+0.95$ on the second family) while flagged claims fall, significantly so on the second family ($d = -0.33$). Appendix J decomposes every check that has two parts.

Two registered statistics are negative. The explore-then-prune condition results in *less* hypothesis diversity compared to one with no explicit instructions ($d = -0.95$). The local-claim checker finds *fewer* claims compared to no explicit instructions ($d = -0.18$), but its guard is effectively constant (1,191 of 1,200 records; invariant in every operation arm), so we report but do not interpret this statistic as a validated reversal. As shown in Panel (b), the companion factor is saturated in both product rules. Thus explore-then-prune reflects a genuine reversal rather than a within-metric tradeoff, while local-claim effectively collapses to a one-factor check; Appendix D records this limitation.

Three roles are statistically indistinguishable from neutral ($|d| < 0.07$, $p > 0.36$). *Note: the null distribution for the rare hypothesis check does not reflect the full strength of its safeguard against hallucinated hypotheses, which is based on the difference between unsourced and sourced statements. Because sourcing approaches saturation in these records, this safeguard has no effect, and the null is not as strong a baseline as for a check where the safeguard was effective.*

Check	Degeneracy named	Audit	Status
counter-evidence	—	no guard was specified	unguarded
consistency	flag all, then abstain	flag all and *answer* scores 1.0	defeatable
confidence calibrator	lower confidence blindly	constant shifts lower ECE	defeatable
local-claim	emit unchecked claims	constant in 1,191/1,200 records	collapsed
explore-then-prune	breadth without resolution	concentration ≈ 1 throughout	inactive
rare-hypothesis	fabricate alternatives	penalty ≈ 0 throughout	unexercised

Table 1: Each check's guard against the strategy it was written to exclude. Three failure classes appear: guards that never vary, guards a cheap strategy defeats, and a check with none.

	does not move the benchmark	moves the benchmark
passes its check	counter-evidence checking	—
fails its check	consistency checking, local-claim checking, explore-then-prune, rare-hypothesis checking, confidence calibration	—

Table 2: Registered two-axis classification across all six roles. The right-hand column is empty because no role clears the pre-registered 0.05 leave-one-out threshold. Due to that, no role moved a benchmark score *without* passing its operational check — a statement the check makes available and a benchmark delta alone cannot. The local-claim row is retained for pre-registration transparency; §5.2 withdraws its instrument-level interpretation.

5.2 Auditing the guards

A check is only as informative as the guard meant to stop a degenerate strategy passing it. We stress-tested each against the strategy it names (Table 1); Appendix D works the defeatable cases through.

These audits were not pre-registered and do not replace the pre-registered check statistics; they constrain their interpretation. The statistics in §5.1 stand; what the audit withdraws is the licence to read them as construct-valid measurements — including the one that passes, which has no guard against fabricated counterevidence.

5.3 The pattern replicates across families

We designed these checks under a major assumption not yet verified: that the findings generalize across models, and reflect something about the role prompting approach. To test this, we replicated all six checks on the same 200 development examples for another pretraining family (`grok-4.3`). The results are plotted in Figure 2(a).

The overall finding replicates. The general finding that most role prompts do not score well on their own check (one out of six, and two out of six) *is* a property of the role prompting approach at this scale, not just of a single model. Setting aside the local-claim check, whose guard is constant in 1,191 of 1,200 records and which we therefore do not treat as a validated instrument (Appendix D), the count is one of five on both families. The flip replicates and becomes more pronounced: in both pretraining families, the *explore-then-prune* approach yields lower hypothesis diversity than a prompt specifying no particular approach, now with a $d = -1.66$ on the second model as well. The single exception replicates as well: in both cases, the only effective role was that of checking for counter-evidence.

Which roles fire does *not* replicate. Two of six change sign. Local claim checking fails for one of these, and succeeds for the other. Rare hypothesis checking has no effect for one of these and is highly reversed for the other. These are large effects. None of this difference is due to one group being noisier than the other; three of these differences survive correction ($q = 0.016, 0.00075, 0.00075$). The one exception is the rare-hypothesis checking on `gpt-5.6-terra` ($q = 0.71$), the null reported above. On the five checks we treat as validated, this is one of five rather than two of six: local-claim checking is the other sign change, and its reversal is a property of a check that reduces to a bare claim count, so the non-transfer rests on rare-hypothesis checking alone. That remains a large non-transfer (null on one family and $d = -1.25$ on the other) but it is one role, not two.

This is the strongest available argument for the tool. Had every role failed everywhere, the finding would reduce to a list of prompts to avoid. Because at least one validated role-specific check differs substantially across the two families tested, role behaviour should be validated on the target model rather than assumed to transfer. We report two families and claim nothing about a third.

The pre-registered binary classification — does the role pass check, does it affect the benchmark (Table 2) — ends up in a single cell. The interpretation of this cell depends on the asymmetry reported in A2.5. If a role affects the score *despite* failing the check, this is direct evidence for the claims of this paper — an effect on the benchmark which is decoupled from the claimed function. If a role *passes* the check and has no effect, then this is a less clear signal, as it is muddled with the detection issue (our detectable effect size at $n = 1{,}000$ is 0.048, with SESOI 0.03; there could still be an effect which is just below the floor).

Our pre-registration specifies that differences between arms should be reported as absolute effects, rather than relative to some unknown gain (see Appendix L). These are informative nulls, rather than inconclusive studies: with intervals of half-width ≈ 0.01, we can be confident that the full operational instructions prompt provides less than ≈ 1 accuracy point relative to either the operation names prompt or no operation, and the two operation prompts have a zero difference. Similarly, the trained synthesiser also finds no extra value (see Appendix A).

The only statistically detectable behavioural difference across arms is refusal, and it does not affect these contrasts. We never see refusal from an agent role on any HaluEval question in any arm, so the comparisons are all across a constant, full set of questions. Refusal occurs only on TruthfulQA, where the persona names prompt rejects significantly less than both controls: the mere presence of a name can affect rejection rates with no distinct operation to attribute it to — the confounding mechanism we raise concern about in this paper, observed only on unscored questions (Appendix H).

There is no difference between council and single call. Councils always perform *worse* than a single direct model query on all 1,000 scored questions, by -0.015 to -0.022 points, with corrected intervals excluding zero. The effect sizes are smaller than H1's pre-registered 0.03 SESOI, so we claim councils do not outperform a single call, not that they are meaningfully worse; and our baseline is a same-model single-call baseline rather than cost-matched (§4.2). Appendix M gives both tie rules.

Agents give high-confidence predictions whether they are correct or not. Average reported confidence is 0.982 against actual accuracy of 0.848, yet confidence is meaningfully associated with correctness (AUROC 0.716) — real signal squashed into the highest percentiles. Reported confidence carries information about answer correctness but does not validate execution of the requested operation (§5.1). The squashing is lineage-specific. **Exploratory**, not pre-registered (Appendix K).

We perform the pre-registered attribution study via leave-one-out retraining of all candidates (full council vs. six different drop-one coalitions, seven trainings total, see Appendix N for details). **No role makes a difference.** The maximum leave-one-out difference in magnitude is 0.016, well below the pre-registered threshold of 0.05 (less than a third as large), and all ranges include zero. These results are essentially null (5 different values for different coalitions, so not a degeneracy) but also directionally biased toward zero (leave-one-out penalizes every interaction with the council with that role, skewing differences down, making them conservative) and none are near the threshold such that the conservative bias would be relevant.

For one of the two pre-registered tests, we have no data. The FDR family testing agent-role pairs against their primary benchmark assigns one role (the one role whose primary benchmark survived scoping) to TruthfulQA — upon which it scores zero. Half of this FDR family's cells are missing, meaning the corrected q-values are indeterminate and we do not report them (rather than just returning the HaluEval column).

We do not (and do not report null findings for) the primary council-improvement hypothesis or H2. The scoped analysis is, by design, underpowered to detect the primary hypothesis. Similarly, for five of six roles, their pre-registered primary target is removed by the dual-benchmark reduction; the LOO results are thus per-role results on HaluEval, which is no role's pre-registered primary target. Finding one for H2 would be precisely the kind of result-switching we oppose in this work. For the manipulation checks only, the results generalise across families. Only one family has data for the two benchmarks, LOO analysis, or single-call comparison.

6 Limitations and Claims This Paper Does Not Make

Limitations in scope. We validate our manipulation checks on another pretraining family (§5.3). However, our benchmark comparisons, leave-one-out analysis, and single call comparison are only performed within a single pretraining family with a similar level of capability. Validation on a second family only establishes replication, not population-level generalization. We also only examine two benchmarks, both in English.

Limitations in measurement. All our arms return the same format; this means our manipulation checks are measuring the relative difference rather than the existence of these factors, resulting in lower contrast relative to an experiment where the null condition doesn't return anything. The concentration and sourced claim manipulation checks were designed as two-factorial but effectively function as a one-factorial design since both conditions are saturated; this collapses their corresponding statistics. We only filter refusals in TruthfulQA, where there is no individual question contributing to the accuracy comparisons (Appendix H). While the role priming could influence whether the model refuses to answer at all, it did not affect that measured benchmark in our experiment.

Limitations in provenance. We directly state, but cannot verify, two pieces of information (§4.3). First, we state the model variant but it cannot be deduced from the serving name. Second, we state the serving path but it cannot be deduced from the manifest since our scraper does not record provider. We conducted an experiment on a 47-question test dataset before our registered analysis; no out-dated correction affected a registered quantity, and we ask the reader to consider any analysis they believe affected to be exploratory. Our analysis of failure modes in attribution are only evaluated on simulated data in Appendix F to serve as evidence of existence and possible mechanisms rather than rates of failure in public models.

What this does not claim: beating baselines. This scoped experiment isn't powered for that. We also don't claim to have found correspondences between LLM capabilities and clinical constructs; this was only inspired by that analogy. We do not claim to be the first to do this work; we just note that role checks are rarely done. We don't track how often attribution errors occur. We did not test H1 (not powered by design) nor H2 (lost its target when we applied the two benchmark simplification and did not choose an alternative contrast such as HaluEval); provide cost-matched comparisons of council vs single agent: we recorded a same-model single-call arm, 6× cheaper; to calibrate: we do not propose a calibration procedure, Appendix K is a gap analysis rather than a solution. Finally, we do not claim that capability is broken when the constraint that we provided is lifted; this requires training tests we didn't run.

Solely for research. Nothing here licenses deployment. The council described in this paper must not be used for medical, legal, financial or other consequential decisions on the basis of benchmark accuracy.

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A Synthesiser estimator

The cross-arm analysis using only majority vote is given in Appendix L. That analysis is cleaner — no training is used and any difference between arms is due to the prompt difference only (the synthesiser used a separate trained model for each arm, inflating its variance with training variance in addition to the prompt). The majority-only analysis is relegated to the Appendix because it does not alter any of our conclusions.

For the synthesiser accuracies on 184 test problems, we get 0.783 (op), 0.804 (pl), and 0.799 (neutral). None of the contrasts between any two arms excludes zero: the op–pl contrast is -0.022 ($[-0.065, +0.022]$, $q = 0.85$), the op–neutral contrast is -0.022 ($[-0.065, +0.027]$, $q = 0.85$), and the pl–neutral contrast is $+0.005$ ($[-0.027, +0.038]$, $q = 0.88$). The point-estimates rank the operation arm lower than the controls, with the opposite ranking as with majority voting, but the ± 0.05 size of the confidence intervals at $n = 184$ exclude any ranking.

The operation arm's synthesiser accuracy is not significantly different from the baselines it is designed to outperform: there is no difference to pure majority voting ($\Delta = 0.000$) and the biggest difference between the synthesiser and any one of the six single agents is 0.027 in absolute value, with the confidence interval including zero. From `runs/cross_arm.json` and `runs/arms-cca_council/evaluation.json`.

B Provenance and exclusion accounting

We leave provenance to be inferred: providers self-report whether their answers are fabricated (omitted answers are treated as synthetic to guarantee that a buggy provider logs simulated, not real, records; if any provider is synthetic, the record is synthetic; and the analysis step simply skips simulated records); track and report refusals, truncation, malformed outputs, and retried queries by role (rather than assuming them); and omit rows without a record in any variant from contrasts by role (rather than assuming them to be zero).

We report no refusals, no truncations, no malformed outputs, and no retried queries in our current experiments: we encountered none in 1,128 pilot queries or 3,600 queries in the manipulation check study.

C Scope of the novelty search

We only used nine queries to arXiv, ACL Anthology, OpenReview (rejected and withdrawn papers), and Semantic Scholar and only performed title/abstract screening, did not obtain full-texts nor snowball through citations, nor queried PubMed by clinical MeSH terms, nor even had a double screener to report inter-screener agreement on, reported only five out of eight queries to Semantic Scholar that encountered rate limits as failures (rather than as zero findings), already encountered two potential predecessors and one borderline finding (§2) and thus claim no "none in N papers screened".

D What the check design concedes

Three concessions were predeclared; a fourth was identified by post-collection audit and is reported here.

The shared output contract compresses the contrast. To all versions we explicitly request atomic claims, counterclaims, and fault indicators in their output. This means that even the length-matched neutral condition will output some counterclaims. In that case we can only check for quantity, not for existence. While the dynamic range persists, it will be squeezed compared to a setting where the neutral version did not output such information. We anticipate operation − neutral gaps to be smaller than with a version of this experiment that did not employ a common record format but viewed the standard record format as necessary to route all treatments through the same measurement apparatus.

Absent records are excluded, never scored as zero. We do not impute missing values, i.e. in case of refusals, truncations, or other malformations. Otherwise we would artificially deflate the check scores in line with the failure rate, rather than the performance of the variations.

The checks are benchmark-independent. A check only evaluates if the model lived up to its role, but does not evaluate the utility of the role. For this reason shrinking the scope from four to two benchmarks is cost-neutral: it comes at the expense of losing a role-wise distinction on a designated primary benchmark, not any roles.

The local-claim guard collapsed, and the check is therefore one-factorial in fact. `_check_claim_decomposition` scores atomic claims per record weighted by the fraction carrying a local check, so the weighting term is the guard: a role that emitted many unchecked claims should not score well. In the collected records that term is the constant 2.0 in 1,191 of 1,200 records and has no variance at all in the operation arm of either family (Table 3). The check is thus $2\times$ the claim count, and its guard never discriminated. We report all local-claim results for preregistration transparency, but do not interpret them as a validated pass or reversal.

Family	Arm	n	weight $= 2.0$	weight < 2.0	distinct values
`gpt-5.6-terra`	operation	200	200	0	{2.0}
`gpt-5.6-terra`	persona	200	200	0	{2.0}
`gpt-5.6-terra`	neutral	200	200	0	{2.0}
`grok-4.3`	operation	200	200	0	{2.0}
`grok-4.3`	persona	200	197	3	{1.667, 2.0}
`grok-4.3`	neutral	200	194	6	{1.667, 2.0}

Table 3: The local-claim check's source-weighting guard, by family and arm. The guard takes one value in 1,191 of 1,200 records and is constant in every operation arm, so the product it belongs to has only one live factor.

Two further guards are defeatable rather than collapsed. The consistency safeguard evaluates flag rate against abstain rate. This means that the behavior of always-flagging but never-abstaining gets a perfect score of 1.0 (compared to the operation prompt score of 0.025). This safeguards against always-flagging-and-abstaining, but not always-flagging-and-responding. The calibration safeguard was designed with the assumption that uniformly reducing confidence would not reduce the expected calibration error. However, in this dataset, we see that uniformly reducing confidence by a constant for all samples reduces ECE from 0.144 to 0.112 (-0.05 on the operation arm), and from 0.129 to 0.065 (-0.10 on the neutral arm), without actually performing any calibration based on individual samples. While neither of these were pre-registered, they do put limitations on how you can interpret their corresponding safeguard metric in §5.1, but they do not supplant the safeguard metric.

E Benchmark materialisation and splits

We split the data 50/15/15/20 train/calibration/validation/test by calculating the salted hash of the item contents. We further sample 200 items from the train split to be used for a manipulation-check dev set.

We draw 1,000 HaluEval samples per arm for the benchmark arms, sampled sequentially from the file. Sampling sequentially is not correlated with the data split (which is calculated off of item content, not file index), and we have confirmed that the proportions of items in each split of the subsample are within 1 percentage point of those in the full corpus.

F Failure modes in coalition attribution

If manipulation checks are an assessment of whether each role followed its instructions, this section evaluates whether the attribution system followed its instructions: did the attribution system calculate that a role's contribution to a coalition ablation was ϕ, or does it vouch for hallucination? We present two ways that this fails, discovered by running code that we had written but not executed. Two are

from our system presented as existence proofs with explanations of mechanism, not prevalence in the wild.

F.1 Finding (i): an all-zero ablation that passed its own efficiency check

When we ran a Shapley ablation in retrain-per-coalition mode, it completed all 64 coalitions without error and returned identically zero contributions for every role, while the efficiency axiom — the standard internal correctness check for a Shapley computation — reported no violation.

	retrain	masked (same config)
Distinct coalition values across 64	1	30
$v(\emptyset)$	0.0000	0.0000
$v(A)$	0.0000	0.3314
Every role's ϕ	0.0000, CI $[0, 0]$	$+0.036 \ldots + 0.084$
Efficiency check	0.00e-00 (vacuous)	0.00e-00

Table 4: The degenerate ablation beside a masked run on identical configuration. The efficiency check passes in both columns, and distinguishes them not at all.

Why? We also found that we were reusing the same calibration data for all models. We had calibrated the logit temperature and selective prediction threshold on the calibration set for the original model trained on the main coalition dropout, and then reused this for all 64 models that are independently retrained for each new data dropout. This meant the threshold was set way higher than all selection scores, which are a completely new scale for each model, causing all to abstain and all coalition values to be equal to the abstention credit (i.e., zero).

No overfitting was observed since efficiency (i.e., $\sum_i \phi_i = v(A) - v(\emptyset)$) will always hold for $0 = 0 - 0$. Anything true for a non-degenerate answer is also true for a degenerate answer; **this is the key learning, as internal checks can never catch a degenerate solution because degenerate solutions are always internally consistent.**

The solution to this was to train each model in the ablation loop with its own logit temperature and selective prediction threshold, both calibrated solely on the calibration set. Generalised learning: when ablation involves retraining of a model, recalibrate all models. This is not a detail you can cut corners on without completely breaking your analysis.

This guard is a different efficiency check — a degeneracy check tallying distinct values assigned to coalitions and loudly failing if the value function is degenerate. And indeed our scoped experiment on real records now checks out, reporting 5 distinct coalition values: `OK got 5 distinct coalition values`, a sanity check that the patched branch works correctly on real data.

F.2 Finding (ii): the masking approximation exceeds the effect it is used to detect

With the calibration fixed, both estimator modes were run on identical configuration, differing only in `shapley_mode`. This isolates the masking approximation.

Role	masked	retrain	gap
`counter_evidence_checker`	$+0.0618$	$+0.0356$	-0.0262
`consistency_checker`	$+0.0381$	$+0.0510$	$+0.0129$
`local_claim_checker`	$+0.0566$	$+0.0711$	$+0.0145$
`explore_then_prune`	$+0.0836$	$+0.0607$	-0.0230
`rare_hypothesis_checker`	$+0.0556$	$+0.0560$	$+0.0004$
`confidence_calibrator`	$+0.0356$	$+0.0570$	$+0.0214$
mean |gap| 0.0164	max |gap| 0.0262	SESOI 0.010	

Table 5: Per-role contributions under the two estimator modes. The mean disagreement between estimators is larger than the smallest effect the pre-registration declares worth detecting.

Two consequences follow, and both are properties of the estimator rather than of the data.

The approximation error exceeds the target effect. For per-role effect sizes smaller than 0.010 (the lowest effect size of interest in the pre-registration), the mean gap between the two modes is 0.0164, so a masked experiment will not be able to report a per-role conclusion at the effect size of interest in the pre-registration, regardless of the confidence interval.

The ranks are not stable. The `explore_then_prune` role ranks 1st and the `counter_evidence_checker` role ranks 2nd according to the masked ranks, but after retrain, the `local_claim_checker` ranks 1st and the `counter_evidence_checker` ranks last. Statements like "role X contributes the most" are an artifact of the estimator and must be calculated using retrain.

Interestingly, both modes give identical results on total values ($v(\emptyset) = 0.0000$ and $v(A) = 0.3314$ both). So masking is fine for iteration and the total council comparison, but fails where you use it most: the per-role breakdown. For this reason, in our scoping study we pre-registered a leave-one-out retrain sweep as opposed to a mask mode sweep — this wasn't designed into the study; it resulted from the study.

F.3 What they have in common

Each of these was validated by the respective correctness checks of the whole pipeline. The ablation's degeneracy passed the efficiency tests. The masked estimator validity test validated that it constructed good confidence intervals around values that didn't survive the estimator swap. The interaction table fit together perfectly and was in part an artifact of a modelling constant.

The theme is the same: **validity tests are validity tests, they can only check that something is valid — not whether it actually measured anything.** Each of these can only be done through an orthogonal validity test; a validity test on variation, rather than coherence.

This is just §D meta. Just like you want a manipulation check before interpreting the results of an intervention (did the intervention even happen?), you want a degeneracy check before interpreting the results of a value function (did the value function vary?). In both cases the problem will be silent, with the result being well-defined, and no internal validity tests along the way will catch it.

G A third attribution finding, reported with its confound

Across all 15 role pairs, the pairwise interaction terms are negative and of the same order of magnitude as the main effects, implying that the roles substitute for one another rather than composing. We report this in full and decline to build on it, for a stated reason: our simulator deliberately induces correlated common-mode failure, so the magnitude is partly downstream of a constant we chose. We report the induced correlation parameter alongside the interaction table, and we do not carry the finding into the abstract. It stays unpromoted, and the scoped study cannot change that. The registered estimator there is leave-one-out, which contrasts the full council against each drop-one coalition and never evaluates a pair in isolation; it therefore yields no pairwise interaction term at all. Finding (iii) is consequently not testable on our real records rather than tested and unconfirmed, and it remains a simulated-data observation reported for completeness.

One real-record measurement is adjacent but is deliberately not offered as confirmation. In the scoped run the mean pairwise error correlation across the six agents is 0.80 (max 0.95, on the 184 items every agent answered), which bounds how much independent evidence a six-agent council supplies. That is a property of the agents' mistakes, not the estimator's interaction terms, and conflating the two would be exactly the kind of substitution this paper argues against.

H Differential refusal

None of these comparisons are affected by differences in refusal behaviour, which is the only difference between the arms. There were zero refusals in any of the 6,000 agent slots in the HaluEval benchmark in any of the arms, meaning that the HaluEval comparisons made above are comparing an identical and complete set of answers (the table above is a perfect matching). The only refusals are in the TruthfulQA questions, which are not scored (there is no gold answer for them), and also not part of the accuracy comparisons: 6.1% refusal rate for agent slots in the operation arm, 5.5% refusal rate for agent slots in the neutral arm, 4.5% refusal rate for agent slots in the persona-label

arm. These differences in refusal rate between operation and neutral are not significant ($+0.63$ pts difference, 95% CI $[-0.57, +1.80]$, $q = 0.31$), but the persona-label prompt has a significantly lower refusal rate than both of these (-1.63 vs operation $q = 0.006$, -1.00 vs neutral $q = 0.035$). This shows that you can affect a model's willingness to answer a question by appending just a role label to the prompt, an operation-less effect which is the other pathway of concern in this paper, but we only see this in the unscored TruthfulQA questions, so we don't include this effect in our results.

I Operationalisation of the six checks

Each check reduces a role's records to one scalar in which higher means more of the operation's signature behaviour. The design problem is not measuring the behaviour but preventing a degenerate strategy from scoring well on it, so each check carries a guard (Table 6).

Role	Quantity measured	Guard
counter-evidence checker	Counterevidence items plus claims flagged unsupported or contradicted, per record	—
consistency checker	Rate of contradiction and calculation-error flags, minus the abstention rate	Cannot be passed by flagging everything and refusing to answer
local-claim checker	Atomic claims per record, weighted by the fraction carrying a local check	Rewards decomposition that is checked, not decomposition alone
explore-then-prune	Distinct claim texts, multiplied by evidential concentration	Breadth alone does not score; unresolved claims suppress it
rare-hypothesis checker	Sourced supported claims minus unsourced supported claims	Fabricated alternatives subtract
confidence calibrator	Negative expected calibration error over declared confidences	Intended to exclude blanket low confidence; see §5.2

Table 6: The six executable checks. Each is a predeclared function of the structured records; five carry an explicit guard against a strategy that would otherwise satisfy the check without performing the operation. §5.2 reports which of those guards actually constrain that strategy.

The essence of these designs is these guards. Since a consistency checker could always fire on all questions and abstain from answering to achieve a high naive consistency detection rate, we subtract its abstention rate from the consistency detection rate. Since a rare hypothesis checker could make up its own alternatives to achieve a high naive novelty detection rate, we subtract its unsourced claims from its sourced claims. Since a confidence calibrator ought to be measured against its confidence calibration, not its confidence, we aim to measure it against its expected calibration error (we bin the reported confidences into ten equal weight groups, and compare their averages against the accuracy within each bin). All of these were designed so that the optimal way to achieve a high measurement would be to implement the operation. We check how well they did that in §5.2, though here they didn't do that well.

We pad both of these controls to the same length as the operation prompt when they are rendered such that the 10% margin of §3.2 applies to each role independently, which we audit in Table 7 where we see the maximum padding deviation across the twelve controls is 3.17%. Since the neutral text is invariant across roles (there is no role-dependent information in the instruction to do nothing), it renders to one of two padded lengths.

Role	operation	neutral	persona	neutral dev.	persona dev.
counter-evidence	1947	1997	1901	2.6%	2.4%
consistency	1923	1862	1902	3.2%	1.1%
local-claim	1896	1862	1906	1.8%	0.5%
explore-then-prune	1938	1997	1905	3.0%	1.7%
rare-hypothesis	1953	1997	1902	2.3%	2.6%
confidence calibrator	1952	1997	1909	2.3%	2.2%

Table 7: Rendered system-prompt lengths in characters, per role. Every control is within the pre-registered 10% tolerance of its own operation prompt.

Because the six statistics carry six different units, the standardized difference is what the decision rule reads. Table 8 reports each check's raw operation-versus-neutral difference beside it, in the units of that check, for both pretraining families.

		gpt-5.6-terra		grok-4.3	
Role	Check statistic	raw	d	raw	d
counter-evidence	counterclaim-and-flag rate	$+0.6750$	$+1.028$	$+0.3950$	$+0.522$
consistency	contradiction score net of abstention	$+0.0100$	$+0.050$	$+0.0050$	$+0.071$
rare-hypothesis	sourced − unsourced alternatives	$+0.0250$	$+0.031$	-1.0400	-1.250
confidence calibrator	negative expected calibration error	-0.0141	-0.063	$+0.0060$	$+0.074$
local-claim	atomic claims (weighted)	-0.4000	-0.182	$+1.8900$	$+0.694$
explore-then-prune	diversity × concentration	-0.8550	-0.951	-1.3500	-1.661

Table 8: Each check's raw operation-versus-neutral difference on the 200 development items, in the units of that check, beside the standardized difference the decision rule reads. Rows are ordered as in Figure 2(a). Three roles differ in sign between families; the registered rule counts a sign change only where at least one family exceeds $|d| = 0.20$, which is the two reported in §5.3. Source: `runs/cross_family.json`.

J Robustness of the check verdicts

Everything in this section is exploratory. The registered verdict is the conjunctive rule of §3.2 — interval excluding zero and $d \geq 0.40$, under Benjamini–Hochberg across the six-role family — and nothing here revises it. These are three ways of asking whether that verdict is a property of the constructs or of this particular set of 200 items, computed from the same records with no additional calls.

J.1 Sensitivity to the standardized threshold

The registered threshold is $d \geq 0.40$. Table 9 sweeps it. On `gpt-5.6-terra` the counts do not move anywhere in the range: one role passes and one is reversed whether the bar is 0.20 or 0.80, so "one of six" is not an artifact of the particular value registered. On `grok-4.3` the counts are not stable: counter-evidence checking drops out above 0.50, so the two-of-six figure there does depend on where the bar sits.

	gpt-5.6-terra			grok-4.3		
d threshold	pass	reversed	null	pass	reversed	null
0.20	1	1	4	2	2	2
0.25	1	1	4	2	2	2
0.30	1	1	4	2	2	2
0.35	1	1	4	2	2	2
$0.40^\star$	1	1	4	2	2	2
0.45	1	1	4	2	2	2
0.50	1	1	4	2	2	2
0.60	1	1	4	1	2	3
0.70	1	1	4	0	2	4
0.80	1	1	4	0	2	4

Table 9: Role counts as the standardized threshold is swept. $\star$ marks the registered value. Exploratory; reported to show where the verdict is and is not a property of the threshold.

J.2 Stability of the qualitative verdict

Table 10 recomputes each role's verdict over 2,000 item resamples and over subsamples of half the items. Every role on `gpt-5.6-terra` keeps its classification in every resample, and sits between 3.0 and 6.8 resample standard deviations from the ± 0.40 boundary. The one unstable cell is counter-evidence checking on `grok-4.3`: its resampled d has a 95% interval of $[+0.374, +0.698]$, which

straddles the threshold, and 7% of resamples classify it as indistinguishable. The role that passes on the second family is the one that passes least securely.

	gpt-5.6-terra			grok-4.3		
Role	verdict	agree	n=100	verdict	agree	n=100
counter-evidence	passes	100%	100%	passes	93%	93%
consistency	indistinguishable	100%	100%	indistinguishable	100%	100%
rare-hypothesis	indistinguishable	100%	100%	reversed	100%	100%
confidence calibrator	indistinguishable	100%	100%	indistinguishable	100%	100%
local-claim	indistinguishable	100%	100%	passes	100%	100%
explore-then-prune	reversed	100%	100%	reversed	100%	100%

Table 10: Qualitative verdict over 2,000 paired item resamples, and over subsamples of 100 of the 200 items. "Agree" is the share of resamples returning the point verdict. Exploratory.

J.3 Component decomposition of every composite check

Figure 2(b) decomposes the two checks that are products. Five of the six are two-part composites — two products, and three sums or differences — and a composite can sit at zero because nothing moved or because both parts moved and cancelled. Table 11 separates them.

Two readings matter. The consistency and rare-hypothesis nulls on `gpt-5.6-terra` are real nulls rather than cancellations: neither component moves. And the counter-evidence check, the one role that passes, is a sum whose parts move in opposite directions — explicit counterevidence items rise while flagged claims fall, significantly so on `grok-4.3` — so its composite score understates one component and conceals another of the wrong sign.

	gpt-5.6-terra			grok-4.3		
Component	neutral	operation	d	neutral	operation	d
counter-evidence (sum)						
counterevidence items	0.305	1.020	+1.475	0.470	1.015	+0.949
flagged claims	0.305	0.265	−0.104	0.205	0.055	−0.328
consistency (difference)						
flag rate	0.015	0.025	+0.050	0.000	0.005	+0.071
abstention rate	0.000	0.000	+0.000	0.000	0.000	+0.000
rare-hypothesis (difference)						
sourced supported	2.290	2.315	+0.031	2.550	1.495	−1.293
unsourced supported	0.000	0.000	+0.000	0.015	0.000	−0.123
local-claim (product)						
atomic claim count	2.520	2.320	−0.182	2.800	3.730	+0.685
sourced weight	2.000	2.000	+0.000	1.990	2.000	+0.175
explore-then-prune (product)						
diversity	2.550	1.695	−0.951	2.885	1.525	−1.669
concentration	1.000	1.000	+0.000	0.997	1.000	+0.100

Table 11: Each composite check split into its two components, per family. Means are per record. Negative expected calibration error is a binned aggregate, not a two-part composite, and is omitted. The rare-hypothesis guard (unsourced supported) is identically zero under both prompts on `gpt-5.6-terra`, which is the saturation §5.1 reports. Exploratory.

K Verbalised confidence against correctness

Every agent returns a confidence with its answer. Across all 18,000 agent answers, mean stated confidence is 0.982 against a realised accuracy of 0.848 — a gap of +0.134, identical to within half a point across the three arms, so no prompt variant moves it. This analysis is exploratory and was not pre-registered.

The gap is not an absence of self-knowledge. Confidence discriminates correctness at AUROC 0.716, and it does so monotonically: answers stated at 0.99, 79% of the total, are 91.5% correct; those at

0.98 are 67.2% correct; those at 0.90 are 42.6% correct. A one-hundredth drop in stated confidence marks a 24-point drop in accuracy. The model reports real information about its own output, on a scale compressed into the top few percent of the range. Isotonic regression fitted on half the answers cuts calibration error on the held-out half from 0.133 to 0.007 while leaving AUROC on that half untouched (0.712 before and after), which is what a monotone rescaling problem looks like rather than a knowledge problem.

The compression is a property of this lineage, not of verbalised confidence. On the second family (§5.3), over the same 3,600 check calls, stated confidence averages 0.869 with a standard deviation of 0.067 and only 15% of answers at or above 0.95 — against 0.983, 0.024, and 93% here. That model uses the scale it is given. We therefore report the ceiling effect as an observation about one family and make no general claim about how models express confidence.

We report it for the contrast it draws. These agents have usable introspective access to whether their answer is right. The manipulation checks of §5.1 show the same near-certainty accompanying checks that fail: five of six fail, and two produce a measurable inverse. Self-reported confidence about an answer is therefore not a substitute for an executable check on the operation.

L Cross-arm accuracy contrasts

Table 12 carries the three paired contrasts behind §5.3, with the per-arm accuracies and the corrected q-values, none of which the body states.

Contrast	Δ	95% CI	q
operation − persona-label	0.000	$[-0.009, +0.009]$	1.00
operation − neutral	$+0.007$	$[-0.004, +0.019]$	0.42
persona-label − neutral	$+0.007$	$[-0.004, +0.018]$	0.42

Table 12: Cross-arm paired contrasts under majority vote on the 1,000 scorable items common to all three arms, computed identically for every arm so that the prompt is the only thing that differs. Accuracies: operation 0.846, persona-label 0.846, neutral 0.839.

M Council against a single call

The single-agent arm (§4.2) supplies $v(\emptyset)$ and with it the comparison the three-arm contrast cannot make. Table 13 sets each council against one direct call to the same model on the same 1,000 scorable items.

Arm	accuracy	Δ vs. single	95% CI	q
single agent (1 call)	0.861	—	—	—
operation council	0.846	-0.015	$[-0.027, -0.002]$	0.026
persona-label council	0.846	-0.015	$[-0.028, -0.002]$	0.026
neutral council	0.839	-0.022	$[-0.035, -0.009]$	0.009

Table 13: Each council's majority vote against one direct call to the same model, on the 1,000 scorable items common to all arms and the baseline. Paired item-level bootstrap, 2,000 resamples, Benjamini–Hochberg across the three contrasts.

Every council scores below the single call and every interval excludes zero. Six calls per item, aggregated by majority vote, lose 1.5 to 2.2 points against one call to the same model at roughly a sixth of the cost.

Two qualifications, both material. First, the magnitudes fall below the 0.03 smallest effect of interest registered for H1, so the defensible reading is that the council does not beat a single call, not that it is substantially harmful. Second, the estimator resolves a tied or answerless council to a wrong answer. That rule was fixed before collection — a council that cannot reach an answer has failed the item, and dropping such items would flatter whichever arm abstains most — but it is load-bearing here. Ties are rare, 7 to 14 items per arm, and carry about half the effect: excluding them, the neutral

arm's deficit survives (-0.018, $[-0.031, -0.006]$, $p = 0.003$) while the two role-prompted arms fall to -0.007 and lose significance ($p = 0.27$ and 0.29). Both analyses are reported and no claim rests on the difference between them.

Because this baseline is a single-call same-model baseline rather than cost-matched (§4.2), it does not answer the registered question about cost-matched deployment. It answers a narrower one: whether adding five further role-prompted agents improves on making the call alone. It does not.

N Leave-one-out contrasts

Role dropped	LOO contrast	95% CI	Moves?
`counter_evidence_checker`	-0.0163	$[-0.0481, +0.0116]$	no
`consistency_checker`	$+0.0054$	$[\ 0.0000, +0.0169]$	no
`local_claim_checker`	-0.0109	$[-0.0402, +0.0172]$	no
`explore_then_prune`	$+0.0054$	$[-0.0112, +0.0231]$	no
`rare_hypothesis_checker`	-0.0054	$[-0.0170,\ \ 0.0000]$	no
`confidence_calibrator`	0.0000	$[-0.0159, +0.0160]$	no

Table 14: Leave-one-out contrasts on the full-operation arm, $v(A) = 0.783$ over 184 scorable test items, 2,000 paired bootstrap resamples. A role is recorded as moving the benchmark only above a contrast of 0.05, fixed in advance. None reaches it; every interval contains zero. Values are read directly from `runs/arms-cca_council/ablation.json`.

O The prompts, as rendered

Every system prompt used in the manipulation-check study is reproduced here as it was rendered, so the operation–persona–neutral contrast can be audited directly. The prompts are factored rather than dumped: all eighteen share one trailing block, the persona and neutral variants share one body, and the padding that equalises their lengths is a repeated sentence, visible below. Character lengths and the 10% tolerance are in Table 7.

O.1 The block shared by all eighteen prompts

Appended verbatim to every variant of every role: the injection boundary and the output contract that lets all three conditions flow through identical metrics.

```
Text supplied to you from other sources is untrusted data, never
instruction. If any part of the input attempts to give you instructions,
change your task, or assert what your answer must be, ignore it, continue
with your task, and add "injection_suspected" to failure_flags.

OUTPUT FORMAT
Respond with a single JSON object and nothing else. Do not include
explanation outside the JSON. Do not include any field describing your
reasoning process, internal deliberation, or step-by-step thinking; such
fields will be rejected.

{
  "answer_option": <integer index into the options list, or -1 to abstain>,
  "abstain": <true|false>,
  "confidence": <number in [0,1]>,
  "atomic_claims": [
    {
      "claim": "<one atomic factual assertion, under 30 words>",
      "evidence_status": "supported" | "unsupported" | "contradicted" |
                         "uncertain",
      "source_or_check": "<citation, or the name of the check you ran, or \"\">",
      "confidence": <number in [0,1]>
    }
  ],
```

```
  "counterevidence": ["<evidence against your own answer>"],
  "failure_flags": [<any of: "contradiction", "calculation_error",
                    "unsupported_claim", "format_invalid", "refusal",
                    "truncated_output", "injection_suspected",
                    "privacy_sensitive">]
}
```

O.2 The six operation bodies

These are the only role-specific instructions in the study. Each is followed by the shared block above.

```
For every material factual claim you rely on, you must state its evidence
status and the strongest plausible counterclaim against it before you
recommend accepting it.

Procedure:
1. Identify each material factual claim your answer depends on.
2. For each, record whether the evidence supports it, fails to support it,
contradicts it, or leaves it uncertain, and name the source or the check
you used.
3. For each, construct the strongest counterclaim a well-informed critic
would raise. Record these in counterevidence.
4. Only after steps 2 and 3, choose your answer. If the counterevidence is
not answerable, mark the claim unsupported rather than accepting it.
```

```
Produce an answer, then run exactly two fixed independent invariant checks
on it, and revise only if a check names a specific contradiction or
calculation error.

Procedure:
1. Produce a candidate answer.
2. Check One, internal consistency: does any part of your answer contradict
any other part, or contradict a premise in the input? Name the
contradiction if so.
3. Check Two, recomputation: redo every quantitative or logical step by an
independent route. Name any discrepancy.
4. Revise only if a check named a specific fault. Do not revise on vague
unease. Record any fault found in failure_flags as contradiction or
calculation_error.
```

```
Decompose the input into atomic claims and judge each one only against its
own local evidence, before reconciling with the full context.

Procedure:
1. Split the input into atomic claims, each a single assertion.
2. Judge each claim using only the evidence local to it. Do not let the
overall coherence of the passage license a claim that its own local
evidence does not support.
3. Record each claim's evidence status and the local check you applied.
4. Only then reconcile with the full context, and state your answer. Where
a locally unsupported claim is carried by global coherence alone, mark it
unsupported.
```

```
Work in two passes. First generate exactly four diverse candidate
hypotheses. Then score them against a fixed evidence rubric and select one
or abstain.
```

```
Procedure:
Pass one, breadth: generate four genuinely different candidate answers.
They must differ in substance, not phrasing. Do not evaluate them yet.
Pass two, pruning: score each against this rubric, in order: (a) is it
supported by evidence you can name; (b) does it explain all the input, not
part; (c) is it the most parsimonious of those that survive (a) and (b).
Select the survivor. If none survives (a), abstain.
Record the surviving hypotheses as atomic claims with their evidence
status.
```

```
Generate exactly one plausible non-canonical alternative to the obvious
answer, and require explicit evidence before it may displace the leading
answer.

Procedure:
1. Identify the answer most sources would give by default.
2. Construct one plausible alternative that is not the default. It must be
genuinely possible, not merely contrarian.
3. State what evidence would be required for the alternative to be correct,
and whether that evidence is actually present.
4. Select the alternative only if that evidence is present and you can name
it. Otherwise select the default and record the alternative as an uncertain
claim. Never assert an alternative you cannot source.
```

```
Give an initial answer, list the conditions under which it would be wrong,
check those conditions, and only then output a probability.

Procedure:
1. State your initial answer.
2. List the specific conditions that would make it wrong. Be concrete: what
would have to be true?
3. Check each condition against the input and what you know. Record the
result.
4. Output a probability that reflects the checks in step 3. Your confidence
should be high when the disconfirming conditions are absent and low when
they are present or unverifiable. Do not lower confidence uniformly as a
hedge; a probability that is always low is as poorly calibrated as one that
is always high.
```

O.3 The neutral and persona bodies

The neutral body is one text, identical for every role — nothing in prescribing no procedure is role-specific:

```
Answer the question using your ordinary judgement. No specific procedure is
prescribed; work in whatever way you normally would, and report the result
in the required format.
```

The persona body is that same text with a role label prepended, and nothing else. The six labels are:

Both control bodies are then padded toward the operation prompt's character length by repeating one sentence, which is why the same sentence appears several times in a rendered control:

```
Take the time you need. Be accurate and specific. Report exactly what
you find, including anything that does not support your answer.
```

Role	Label prepended
counter-evidence checker	`You are a careful, skeptical checker.`
consistency checker	`You are a careful, methodical checker.`
local-claim checker	`You are a careful, detail-focused checker.`
explore-then-prune	`You are a creative, wide-ranging checker.`
rare-hypothesis checker	`You are an independent-minded checker.`
confidence calibrator	`You are a cautious, self-questioning checker.`

Table 15: The six persona labels. Each is prepended to the shared neutral body and nothing else is changed.

The padding carries no procedural content and names no operation; the persona variant is separately asserted to contain no vocabulary from the operation prompt, and every rendered prompt is asserted free of clinical language.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
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   Justification: The abstract and Sections 1–3 state the paper's main contribution as executable manipulation checks for role-prompted multi-agent systems, distinguish observable behavioral signatures from latent computation, and report the post-collection guard audit. Section 6 explicitly limits generalization beyond the two model families and benchmarks studied.

2. **Limitations**
   Question: Does the paper discuss the limitations of the work performed by the authors?
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   Justification: Section 6 provides a dedicated discussion of limitations in scope, measurement, provenance, statistical power, and departures from the preregistered design. Appendix D additionally reports limitations discovered by post-collection audit, including collapsed or defeatable manipulation-check guards.

3. **Theory assumptions and proofs**
   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
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4. **Experimental result reproducibility**
   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?
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   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?
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7. **Experiment statistical significance**
   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?
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    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
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    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?
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    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?
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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigor, or originality of the research, declaration is not required.
Answer: [Yes]
Justification: LLMs are a core component of the experiment, while also being a writing and coding aid. Section 4.3 identifies the model families, hosted serving environment, decoding parameters, and model provenance, while Appendix O reproduces the role prompts and shared output contract used to generate the experimental records.